

STANDARD ERRORS OF ESTIMATORS OF THE INTENSITY FUNCTION UNDER THE POISSON HYPOTHESIS

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SUMMARY

Standard errors of estimators of the intensity function of a stationary point process are obtained under the assumption that the process is Poisson. The corresponding estimators for the standard errors may be used as approximations for processes other than Poisson. Monte Carlo experiments carried out to assess the usefulness of the approximations for some well-known processes confirm this. These are discussed in the final section.

1. Introduction

Consider a stationary point process $N(\omega, B)$, $B \in \mathcal{B}$, on a probability space $(\Omega, \mathcal{A}, P)$ and let ν be the mean rate of the process. (For definitions see Steinijans, 1970.) The second order properties of the process are determined by the intensity function, $m(x)$, defined for $x > 0$ by

$$(1) \quad m(x) = \lim_{\Delta x \rightarrow 0^+} \frac{\text{Prob. \{point in } (t+x, t+x+\Delta x) \text{ / point at } t \}}}{\Delta x}$$

The intensity function is simply related to the covariance density (Bartlett, 1955) and if the process is an equilibrium renewal process it is equivalent to the renewal density (Smith, 1958).

To estimate the intensity function one has to use an estimator suitable for the type of observation available. In cases where an observation is available on the process of lengths, an appropriate estimator is given in Cox (1965). For an observation on the process of counts an estimator for the intensity function is given in Linhart (1973).

Standard errors of the estimators mentioned above are given in Linhart (1975) and Linhart (1973) respectively, but the formulae involved are computationally inconvenient. A conditional (given n points have occurred) standard error for the estimator of the intensity function based on lengths under the hypothesis that the process is Poisson, is given in Cox (1965).

2. Estimation from observed lengths

Assume that we have an observation $\{L_0, L_1, \dots, L_n\}$ on the process of lengths, the estimator for the intensity function given in Cox (1965) is

$$(2) \quad \hat{m}(u+\Delta u/2) = \frac{t_0}{n(t_0-u-\Delta u/2)} \sum_{r=1}^n (n-r+1) n f_r(u)$$

$$\text{where } t_0 = \sum_{i=0}^n L_i, \quad n f_r(u) = \frac{1}{\Delta u(n-r+1)} \sum_{i=0}^{n-r+1} Y_i^{(r)}(u)$$

and $Y_i^{(r)}(u) = 1$ if $u < L_i + L_{i+1} \dots + L_{i+r-1} \leq u + \Delta u$, zero otherwise.

To facilitate the calculation of moments of $\hat{m}$ it is convenient to write (2) in the form

$$(3) \quad \hat{m}(u+\Delta u/2) = t_0 S(u, \Delta u) / (\Delta u n(t_0-u-\Delta u/2))$$

$$\text{where } S(u, \Delta u) = \int_0^{t_0-u} \int_{u+x}^{u+\Delta u+x} dN(\omega, y) dN(\omega, x).$$

It is easy to show that $E S(u, \Delta u) = \nu (t_0-u-\Delta u/2) \int_u^{u+\Delta u} m(x) dx$ and hence if we neglect the variation due to $\hat{\nu} = n/t_0$ (whose standard error is of the order ν/t_0) we obtain

$$(4) \quad E \hat{m}(u+\Delta u/2) \simeq \frac{1}{\Delta u} \int_u^{u+\Delta u} m(x) dx,$$

an average of the intensity function near the argument $u+\Delta u/2$.

The variance of $S(u, \Delta u)$ involves a sum of quadruple integrals of the type given in (3) and for a general point process there does not seem to be a convenient expression for this variance. In the case of a Poisson process, however, the integrals do simplify and after considerable manipulation one obtains

$$(5) \quad \text{Var } S(u, \Delta u) = \nu^3 \Delta u^2 (4t_0 - 4u - 3\Delta u/4) + \nu^2 \Delta u (t_0 - u - \Delta u/2).$$

If we again neglect the variation due to $\hat{\nu}$ we obtain

$$(6) \quad \text{Var}(\hat{m}) \simeq \text{Var}(S(u, \Delta u)) / \{(\Delta u)^2 \nu^2 (t_0 - u - \Delta u/2)^2\}.$$

An estimator of $\text{Var}(\hat{m})$ based on (6) is

$$(7) \quad \hat{\text{Var}}(\hat{m}) = \frac{\hat{\nu}(4t_0 - 4u - 3\Delta u/4)}{(t_0 - u - \Delta u/2)^2} + \frac{1}{\Delta u(t_0 - u - \Delta u/2)}$$

In many cases $\Delta u \ll t_0 + u$ and (7) may be approximated by

$$(8) \quad \hat{\text{Var}}(\hat{m}) = (4\hat{\nu} + 1/\Delta u)/(t_0 - u).$$

3. Estimation from counts

Assume that we have an observation $\{K_1, K_2, \dots, K_p\}$ on the process of counts, (i.e. K_i points occurred in the i -th of p consecutive intervals of length τ .) An estimator of the intensity function given in Linhart (1973) is:

$$(9) \quad \hat{m}(r\tau) = pS(r, \tau) / n\tau(p-r),$$

$$\text{where } S(r, \tau) = \sum_{i=1}^{p-r} K_i K_{i+r}, \quad r=1, 2, \dots, p-1 \text{ and } n = \sum_{i=1}^p K_i.$$

If we neglect the variation due to $\hat{\nu} = n/p\tau$ (the standard error of $\hat{\nu}$ is of the order $\nu/p\tau$) the expectation of $\hat{m}$ is given by

$$E \hat{m} \simeq \int_{-\tau}^{\tau} (\tau - |x|) m(r\tau + x) dx / \tau^2,$$

which is again a weighted average of the intensity function near the argument $r\tau$. In this case, however, the window is triangular.

As in the previous case the variance of $S(r, \tau)$ does not conveniently simplify. In the case of a Poisson process, however, it can be shown, using the fact that the Poisson process has independent increments, that

$$(10) \quad \text{Var}[S(r, \tau)] = (p-r)(\nu\tau)^2(1+2\nu\tau) + 2(p-2r)(\nu\tau)^3, \quad r=1, 2, \dots, p-1.$$

If we neglect the variation due to $\hat{\nu}$ we obtain

$$(11) \quad \text{Var}(\hat{m}) \simeq \text{Var}(S(r, \tau)) / \tau^4 \nu^2 (p-r)^2, \quad r=1, 2, \dots, p-1,$$

which can be estimated by

$$(12) \quad \hat{\text{Var}}(\hat{m}) = \frac{(1+2\hat{\nu}\tau)}{\tau^2(p-r)} + \frac{2(p-2r)\hat{\nu}}{(p-r)^2}, \quad r=1, 2, \dots, p-1.$$

In many cases the following approximation is adequate.

$$(13) \quad \hat{\text{Var}}(\hat{m}) = (4\hat{\nu} + 1/\tau) / (p-r), \quad r=1, 2, \dots, p-1.$$

4. Monte Carlo Experiments

Experiments were carried out to assess the usefulness of (7), (8), (12) and (13) as approximations to the standard error of the estimated intensity function for both the Poisson and other point processes. In each case empirical standard errors were computed using 50 independent observations and these were compared with the values given by (8) and (13). Typical results are given in the tables.

Tables 1 to 3 give the empirical standard error (Emp. S.E.), the estimated standard

error (Est. S.E.) using (8) and the percentage relative error (% R.E.) which is $100|\text{Emp. S.E.} - \text{Est. S.E.}|/\text{Emp. S.E.}$. Tables 4 to 6 relate to observed counts. Here the Est. S.E. is calculated using (13).

The point processes used were the Poisson process, (tables 1 and 4), the gamma renewal process (tables 2 and 5), the uniform renewal process (table 3) and the normal autoregressive moving average (ARMA) (1,0) process (table 6). Results for (8) using the normal ARMA process have been reported in Linhart (1975), they are similar to those given here.

It can be seen from the tables that (8) and (13) tend to overestimate the standard error for all processes studied. In the case of the Poisson process this bias is small and is due to the approximations made in the derivations. For other processes the bias is obviously mainly due to the fact that the formulae do not apply for the point processes being observed. The approximations do seem, in all cases considered, to give a reasonable approximation to the order of magnitude of the standard errors, although their performance will depend on the point process operating.

TABLE 1:
Poisson process, $\lambda = 1$

	$u =$	0,5	1,0	1,5	2,0	2,5
$t_0=50$ $\Delta u=0,05$	Emp. S.E.	0,68	0,68	0,70	0,75	0,67
	Est. S.E.	0,70	0,70	0,70	0,71	0,71
	% R.E.	3	3	0	7	6
$t_0=100$ $\Delta u=0,25$	Emp. S.E.	0,34	0,36	0,38	0,39	0,40
	Est. S.E.	0,40	0,40	0,41	0,41	0,41
	% R.E.	18	11	8	5	3

TABLE 2:
Gamma renewal process, $\lambda = 1$; $k = 2$

	$u =$	0,5	1,0	1,5	2,0	2,5
$t_0=50$ $\Delta u=0,1$	Emp. S.E.	0,40	0,47	0,47	0,41	0,50
	Est. S.E.	0,49	0,49	0,50	0,50	0,50
	% R.E.	23	5	6	22	0
$t_0=100$ $\Delta u=0,1$	Emp. S.E.	0,27	0,29	0,29	0,34	0,34
	Est. S.E.	0,34	0,35	0,35	0,35	0,35
	% R.E.	29	20	20	3	3

TABLE 3:
Uniform (0;1) renewal process

	u =	0,5	1,0	1,5	2,0	2,5
$t_0=50$ $\Delta u=0,1$	Emp. S.E.	0,35	0,37	0,46	0,39	0,40
	Est. S.E.	0,60	0,61	0,61	0,61	0,62
	% R.E.	72	64	32	57	54
$t_0=100$ $\Delta u=0,1$	Emp. S.E.	0,25	0,27	0,30	0,24	0,28
	Est. S.E.	0,42	0,43	0,43	0,43	0,43
	% R.E.	70	57	43	78	53

TABLE 4:
Poisson process, $\lambda = 1$

	r =	5	10	15	20	25
$p = 500$ $\tau = 0,1$	Emp. S.E.	0,45	0,50	0,46	0,50	0,46
	Est. S.E.	0,53	0,53	0,54	0,54	0,54
	% R.E.	18	7	17	8	18
$p = 1\ 000$ $\tau = 0,1$	Emp. S.E.	0,32	0,34	0,30	0,35	0,37
	Est. S.E.	0,38	0,38	0,38	0,38	0,38
	% R.E.	18	11	26	8	2

TABLE 5:
Gamma renewal process, $\lambda = 1$; $k = 2$

	r =	5	10	15	20	25
$p = 500$ $\tau = 0,5$	Emp. S.E.	0,093	0,086	0,085	0,096	0,074
	Est. S.E.	0,127	0,128	0,128	0,129	0,130
	% R.E.	37	42	51	34	75
$p = 1\ 000$ $\tau = 0,5$	Emp. S.E.	0,061	0,064	0,052	0,064	0,063
	Est. S.E.	0,090	0,091	0,091	0,091	0,092
	% R.E.	47	40	73	41	44

TABLE 6:
Normal ARMA (1;0) process, $\mu = 10$; $\sigma^2 = 1$; $\phi = 0,5$

	$r =$	5	10	15	20	25
$p = 500$ $\tau = 2,0$	Emp. S.E.	0,028	0,030	0,028	0,025	0,026
	Est. S.E.	0,030	0,030	0,030	0,030	0,030
	% R.E.	7	1	8	22	19
$p = 1\ 000$ $\tau = 2,0$	Emp. S.E.	0,017	0,021	0,024	0,019	0,020
	Est. S.E.	0,021	0,021	0,021	0,021	0,021
	% R.E.	25	2	11	13	7

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